

## **Exp No: 01**

### **Transformation between Cartesian, Cylindrical and spherical coordinate system**

#### **Aim:**

To write program for Transformation between Cartesian, Cylindrical and spherical coordinate system using SCILAB.

#### **Software required:**

- SCILAB version 6.0.1

#### **Program:**

#### **Test case (i) Transformation of Cartesian coordinate points into Cylindrical and spherical coordinate system**

#### **Code:**

```
clc;
x = input('Enter the value of X = ');
y = input('Enter the value of Y = ');
z = input('Enter the value of Z = ');
// Cartesian to Cylindrical
r = sqrt(x^2 + y^2);
phi = atand(y/x);
disp('Cartesian to Cylindrical Coordinates [r phi z] =');
disp([r phi z]);
// Cartesian to Spherical
R = sqrt(x^2 + y^2 + z^2);
theta = acosd(z/R);
phi = atand(y/x);
disp('Cartesian to Spherical Coordinates [R theta phi] =');
disp([R theta phi]);
```

## Output:

```
Scilab 2026.1.0 Console

Enter the value of X = 10

Enter the value of Y = -1

Enter the value of Z = 3

"Cartesian to Cylindrical Coordinates [r phi z] ="
10.049876 -5.7105931 3.
"Cartesian to Spherical Coordinates [R theta phi] ="
10.488088 73.379049 -5.7105931

-->
```

## Practical Calculation:

Case (i):  $x = 10$ ,  $y = -1$ ,  $z = 3$  Cartesian to Cylindrical Coordinates

$$r = \sqrt{x^2 + y^2} = \sqrt{10^2 + (-1)^2} = \sqrt{101} \approx 10.05$$
$$\phi = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{-1}{10}\right) = -5.71$$
$$z = 3$$

Cartesian to Spherical Coordinates

$$R = \sqrt{x^2 + y^2 + z^2} = \sqrt{10^2 + (-1)^2 + 3^2} = \sqrt{110} \approx 10.49$$
$$\theta = \cos^{-1}\left(\frac{z}{R}\right) = \cos^{-1}\left(\frac{3}{10.49}\right) = 73.38$$
$$\phi = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{-1}{10}\right) = -5.71$$

Final Results:

$$[R, \theta, \phi] = [10.49, 73.38, -5.71]$$

## Test case (ii) Transformation of Cylindrical coordinate points into Cartesian and spherical coordinate system

### Code :

```
clc;

clear;

// Input values

r = input('Enter the value of r = ');
```

```

phi = input('Enter the value of phi (in degrees) = ');
z = input('Enter the value of z = ');

// Cylindrical to Cartesian
x = r * cosd(phi);
y = r * sind(phi);
disp('Cylindrical to Cartesian Coordinates [x y z] =');
disp([x y z]);

// Cylindrical to Spherical
R = sqrt(r^2 + z^2);
theta = acosd(z / R);
disp('Cylindrical to Spherical Coordinates [R theta phi] =');
disp([R theta phi]);

```

### Output:

```

Scilab 2026 1.0 Console

Enter the value of r = 4

Enter the value of phi (in degrees) = 35

Enter the value of z = -5

"Cylindrical to Cartesian Coordinates [x y z] ="
3.2766082  2.2943057  -5.
"Cylindrical to Spherical Coordinates [R theta phi] ="
6.4031242  141.34019  35.

--> |

```

### Practical Calculation:

Case (iii)  $\phi = 100^\circ$ ,  $\theta = 45^\circ$ ,  $r = 5$

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

$$x = 5 \sin 45^\circ \cos 100^\circ$$

$$y = 5 \sin 45^\circ \sin 100^\circ$$

$$z = 5 \cos 45^\circ$$

$$x = 5 \times 0.7071 \times (-0.1736)$$

$$y = 5 \times 0.7071 \times 0.9848$$

$$z = 5 \times 0.7071$$

$$x = -0.6139$$

$$y = 3.4818$$

$$z = 3.5355$$

Ex 6: Spherical to Cylindrical

$$\rho = r \sin \theta$$

$$\phi = \phi$$

$$z = r \cos \theta$$

$$\rho = 5 \sin 45^\circ$$

$$z = 5 \cos 45^\circ$$

$$\rho = 5 \times 0.7071$$

$$z = 5 \times 0.7071$$

$$\rho = 3.5355$$

$$z = 3.5355$$

### Test case (iii) Transformation of spherical coordinate points into Cylindrical and Cartesian coordinate system

**Code:**

```
clc; clear;

// Input r = input('Enter the value of r = ');

phi = input('Enter the value of phi (in degrees) = ');

theta = input('Enter the value of theta (in degrees) = ');

// Spherical to Cartesian

x = r * sind(theta) * cosd(phi);

y = r * sind(theta) * sind(phi); z = r * cosd(theta);

disp('Spherical to Cartesian Coordinates [x y z] =');

disp([x y z]);

// Spherical to Cylindrical rho = r * sind(theta);

disp('Spherical to Cylindrical Coordinates [rho phi z] =');

disp([rho phi z]);
```

**Output:**

Enter the value of r = 5

Enter the value of phi (in degrees) = 45

Enter the value of theta (in degrees) = 100

"Spherical to Cartesian Coordinates [x y z] ="  
3.4818212 3.4818212 -0.8682409

"Spherical to Cylindrical Coordinates [rho phi z] ="  
4.9240388 45. -0.8682409

### Practical Calculation:

Case (iii)

$$r = 5, \theta = 45^\circ, \phi = 100^\circ$$

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

$$x = 5 \sin 45^\circ \cos 100^\circ$$

$$y = 5 \sin 45^\circ \sin 100^\circ$$

$$z = 5 \cos 45^\circ$$

$$x = 5 \times 0.7071 \times (-0.1736)$$

$$y = 5 \times 0.7071 \times 0.9848$$

$$z = 5 \times 0.7071$$

$$x = -0.6139$$

$$y = 3.4818$$

$$z = 3.5355$$

Spherical to Cylindrical

$$\rho = r \sin \theta$$

$$\phi = \phi$$

$$z = r \cos \theta$$

$$\rho = 5 \sin 45^\circ$$

$$z = 5 \cos 45^\circ$$

$$\rho = 5 \times 0.7071$$

$$z = 5 \times 0.7071$$

$$\rho = 3.5355$$

$$z = 3.5355$$

### Result:

Thus, the program was verified Transformation between Cartesian and Cylindrical coordinate system its classified two categories

- Cartesian to cylindrical and spherical coordinate system
- Cylindrical to cartesian and spherical coordinate system
- spherical to cartesian and cylindrical coordinate system

using SCILAB was successfully.

| Parameters                                                           | Observed/Simulated Result                                                                                                                            | Theoretical Result                                                                                                                    |
|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Case 1:<br>x = 10,<br>y = -1,<br>z = 3                               | $\rho = 10.049876$ ,<br>$\varphi = 354.289407^\circ$ ,<br>z = 3<br>$r = 10.488088$ ,<br>$\theta = 73.379049^\circ$ ,<br>$\varphi = 354.289407^\circ$ | $\rho = 10.050$ ,<br>$\varphi = 354.289^\circ$ ,<br>z = 3<br>$r = 10.488$ ,<br>$\theta = 73.379^\circ$ ,<br>$\varphi = 354.289^\circ$ |
| Case 2:<br>$\rho = 4$ ,<br>$\varphi = 35^\circ$ ,<br>z = -5          | x = 3.276608,<br>y = 2.294306,<br>z = -5<br>$r = 6.403124$ ,<br>$\varphi = 35^\circ$ ,<br>$\theta = 141.340192^\circ$                                | x = 3.277,<br>y = 2.294,<br>z = -5<br>$r = 6.403$ ,<br>$\varphi = 35^\circ$ ,<br>$\theta = 141.340^\circ$                             |
| Case 3:<br>r = 10,<br>$\varphi = 100^\circ$ ,<br>$\theta = 45^\circ$ | x = -1.227878,<br>y = 6.963642,<br>z = 7.071068<br>$\rho = 7.071068$ ,<br>$\varphi = 100^\circ$ ,<br>z = 7.071068                                    | x = -1.228,<br>y = 6.964,<br>z = 7.071<br>$\rho = 7.071$ ,<br>$\varphi = 100^\circ$ ,<br>z = 7.071                                    |